

# Chinese-SkillSpan: a benchmark for competency span extraction in Chinese job advertisements

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## ABSTRACT

**Background.** Chinese job advertisements express competencies through coordinated phrases and context-dependent terms. Comparing extraction systems requires consistent decisions about which characters form each mention and which competency type it expresses.

**Methods.** We developed Chinese-SkillSpan from 22,840 sentences across four recruitment collections. Character-level labels distinguish language, knowledge-related requirements, occupational skills, and transversal competencies. An ESCO-informed handbook was refined through human coding, disagreement analysis, and adjudication. Human reference labels provide the primary model-evaluation targets. Silver labels support training, checkpoint selection, and supplementary agreement checks against generated annotations. Silver quality checks combine structural validation with sampled human review before and after bulk annotation. Three annotators independently applied the final handbook, with machine and peer labels hidden. We compared prompted models, supervised adaptation, and multilingual encoders.

**Results.** Silver-supervised Qwen adaptation improved typed exact micro-F1 on the human reference from a no-adaptor baseline of 0.361 to  $0.540 \pm 0.035$  (mean  $\pm$  sample SD across three training seeds) under fixed inference rules. Independent blinded annotation achieved mean pairwise typed exact F1 of 0.818 and boundary-only F1 of 0.850. The distinction identifies type assignment as an additional source of annotation difficulty.

**Conclusions.** Chinese-SkillSpan pairs source-text spans with explicit annotation rules and separate human and Silver labels, enabling model comparisons and inspection of boundary and type errors. The human reference supports diagnostic comparisons on development-used data; evaluation on new source documents is needed to establish generalization.

## INTRODUCTION

Online job advertisements describe the knowledge and skills employers seek and provide evidence of changing occupational requirements (Podjanin and Strietska-Ilina, 2020; Senger et al., 2024).

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Document-level methods retrieve skills from predefined inventories (Bhola et al., 2020), whereas span extraction locates the words expressing a requirement (Gnehm et al., 2022). Job Skill Named Entity Recognition (JobSkillNER) identifies these mentions and assigns a type to each span. Pre-serving source wording makes boundary and type assignments inspectable.

Even short Chinese phrases can require several annotation decisions. A technology name can denote tool use in one clause and knowledge of principles in another. Coordination may express separate activities or share an action that must remain attached to its objects. Chinese normally lacks word spaces, so tokenization alone cannot resolve these semantic boundaries (Peng and Dredze, 2015; Chen and Kong, 2021). Similar ambiguities occur in other languages (Nguyen et al., 2024), but dataset translation does not determine which Chinese substrings to select. Explicit context-sensitive rules are needed; Figure 4 pairs Chinese examples with English explanations.

Existing resources provide precedents, but their annotation schemes differ. English SkillSpan permits overlapping SKILL and KNOWLEDGE layers (Zhang et al., 2022b). Kompetenzer adds ESCO-derived fine-grained classification to Danish and English spans (Zhang et al., 2022a). German recruitment work extracts education, experience, and language requirements before finer classification (Gnehm et al., 2022). Document-level retrieval and Chinese skill-demand forecasting use other prediction targets (Bhola et al., 2020; Chen et al., 2024).

We introduce Chinese-SkillSpan, a benchmark for competency span extraction in Chinese job advertisements. Built from a corpus of 22,840 sentences across four recruitment collections, it combines annotation guidelines, distinct supervision and reference subsets, scoring protocols, and model baselines. The handbook distinguishes language (L), knowledge (K), occupational skills (S), and transversal competencies (T). In Chinese-SkillSpan, we use *competency spans* for contiguous source-text mentions across all four categories and *occupational skills* specifically for S. The scheme draws on ESCO’s high-level categories and the knowledge–skill distinction in the European Qualifications Framework (European Commission, 2024; Council of the European Union, 2017), without assigning ESCO concept identifiers. Character offsets preserve each mention’s wording. Recruitment qualifications and explicit industry-background requirements extend the project K category beyond substantive knowledge; their scope is defined separately below.

The study addresses three linked questions: how to delimit and type Chinese competency mentions, how consistently annotators apply those rules, and how extraction models perform under the resulting conventions. The handbook supplies boundary and context rules with positive and negative examples. A separate three-annotator blinded study measures agreement under the final handbook, while historical coding and assisted review document reference construction and Silver quality checks. Matched Qwen adaptation and Chinese-supervised encoder comparisons assess model performance, with diagnostics for type, span length, and empty-reference sentences. Gold150 supplies the primary human targets for diagnostic comparisons; it also informed the guidelines. Silver labels support training, selection, and supplementary generated-target checks. The blinded study assesses handbook agreement using individual annotations, without consensus test labels. We document each subset’s role and availability following guidance on datasets and annotation agreement (Bender and Friedman, 2018; Gebru et al., 2021; Artstein and Poesio, 2008).

## RELATED WORK

**Skill extraction and recruitment datasets.** Recruitment datasets differ in prediction targets. Sayfullina et al. (2018) ask whether a candidate soft-skill phrase refers to an applicant, whereas Bhola et al. (2020) retrieve skills from a predefined inventory for an entire advertisement. Span extraction locates requirement wording. Within that task, SkillSpan annotates English SKILL and KNOWLEDGE in two potentially overlapping layers; hard and soft skills describe its coverage rather than its label inventory (Zhang et al., 2022b). Kompetencer combines manual skill and knowledge spans with ESCO-derived fine-grained labels in Danish and English; its distantly supervised Danish test labels were manually corrected (Zhang et al., 2022a). Gnehm et al. (2022) extract coarse EDU/EXP/LNG spans before refining skill areas for taxonomy matching. Chinese recruitment research includes both forecasting and entity extraction. Job-SDF evaluates skill-demand forecasting at several aggregation levels (Chen et al., 2024). For occupational profiling, Cao et al. (2021) manually annotated skill and specialty requirements using BIOES and trained a BERT–BiLSTM–CRF model. Chinese-SkillSpan instead uses flat character spans with four types and explicit rules for coordination and contextual interpretation. Taxonomy-based weak supervision can reduce manual annotation work (Zhang et al., 2022c), but generated labels still need validation. Nguyen et al. (2024) report difficulties with conjoined and ambiguous mentions across six recruitment datasets. Senger et al. (2024) also identify scarce public annotations and inconsistent terminology. Table 1 compares the prediction units, annotation layers, and available materials of selected resources.

**Table 1.** Selected recruitment resources: prediction units, annotation layers, and reported materials. Table 2 details Chinese-SkillSpan counts and data roles.

| Work / language                        | Prediction unit               | Actual annotation layers                                                                 | Material scope                                                                                |
|----------------------------------------|-------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Sayfullina et al. (2018); English      | Candidate phrase in context   | Relevance to an applicant; binary disambiguation, not open-boundary extraction           | Annotated soft-skill contexts                                                                 |
| Gnehm et al. (2022); German            | Spans, then finer areas       | Coarse EDU/EXP/LNG; finer skill areas and taxonomy matching                              | Article and released domain-adapted models                                                    |
| Zhang et al. (2022b); English          | Source-text spans             | SKILL / KNOWLEDGE; overlap permitted between the two layers                              | Dataset, code, and annotation guidance                                                        |
| Zhang et al. (2022a); Danish / English | Spans and fine-grained labels | Manual skill/knowledge spans; ESCO API distant labels; Danish test-label correction      | Dataset and code                                                                              |
| Cao et al. (2021); Chinese             | Recruitment entities          | Manually annotated skill / specialty entities; BIOES                                     | Article and supporting data files                                                             |
| Chinese-SkillSpan; Chinese             | Flat character spans          | L/K/S/T; frozen human reference and Silver supervision; separate blinded agreement study | Handbook, protocols, baseline predictions, scorer and independent coding exports (Appendix D) |

**Chinese span annotation and competency guidelines.** Chinese NER resources cover social-media text (Peng and Dredze, 2015) and fine-grained entity categories (Xu et al., 2020). Multilingual studies examine recognition in noisy text (Fetahu et al., 2023) and changes in contextual entity types (Luo et al., 2025). These tasks provide background but do not specify how to divide recruitment requirements into LSKT spans. Chinese boundary detection is complicated by the absence of explicit word delimiters (Chen and Kong, 2021); recruitment annotation also distinguishes knowledge, occupational actions, and transversal competencies in each clause. ESCO provides a multilingual vocabulary for labor-market information (le Vrang et al., 2014; European Commission, 2024). SkillLens combines human involvement, skill extraction, and ESCO linking across European job advertisements (De Santo et al., 2026). Chinese-SkillSpan adapts ESCO’s high-level categories to LSKT annotation in Chinese. The handbook specifies when tool use denotes S rather than K, when coordinated objects must retain a shared action, and when an experience qualifier belongs inside the span. Agreement analysis and adjudication consequently address both unitization and categorization (Artstein and Poesio, 2008).

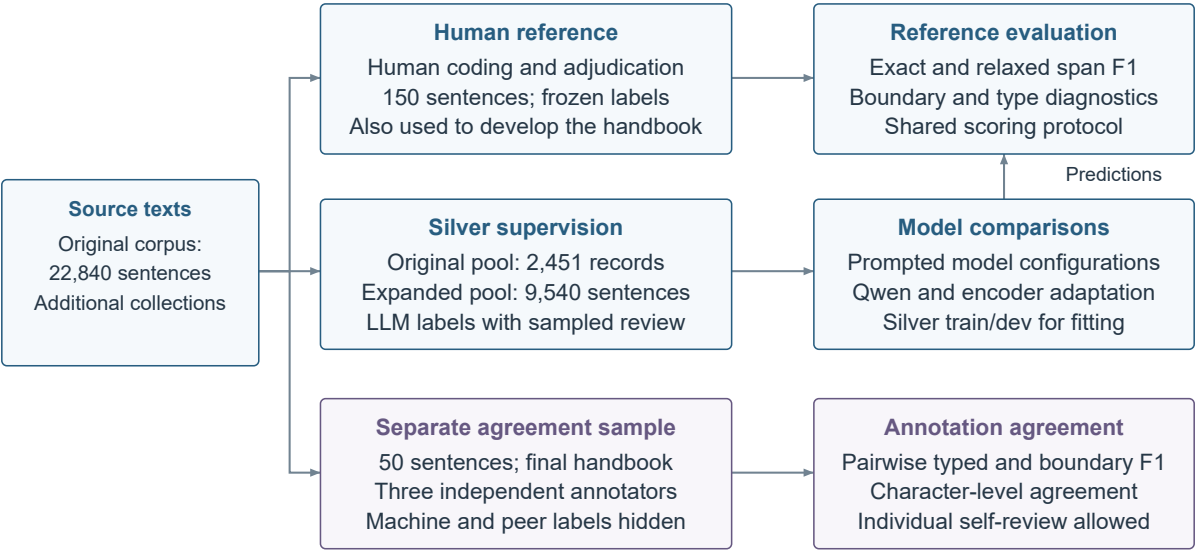
**Model-assisted annotation and extraction models.** LLMs can supply first-pass labels, with their usefulness depending on task instructions and human validation (Ding et al., 2023; Li et al., 2023; Ul Haq et al., 2026). GoLLIE and subsequent guideline-refinement work study how annotation instructions shape model outputs (Sainz et al., 2024; Kim et al., 2026). In recruitment, JobScape generates synthetic advertisements for skill-to-taxonomy matching (Magron et al., 2024). Chinese-SkillSpan retains collected recruitment texts and adds model-generated annotations with documented review. The annotation prompts follow the handbook used at each stage. Encoder models offer one way to learn from these labels. Pretrained encoders and structured decoding (Devlin et al., 2019; Lafferty et al., 2001) can be combined with continued pretraining on recruitment text (Gururangan et al., 2020; Zhang et al., 2022b). SkillSpan introduced the domain-adapted JobBERT and JobSpanBERT baselines (Zhang et al., 2022b). ESCOXML-R uses taxonomy-driven multilingual pretraining (Zhang et al., 2023), and NNOSE retrieves training-data representations to support extraction, including infrequent patterns (Zhang et al., 2024). Reproducing these methods on their original tasks and adapting them to Chinese LSKT extraction require separate protocols because the languages, annotation layers, and splits differ. Other approaches use text-to-structure generation in UIE (Lu et al., 2022), distilled supervision in UniversalNER (Zhou et al., 2024), open-type encoder extraction in GLiNER (Zaratiana et al., 2024), and generative extraction with self-verification in GPT-NER (Wang et al., 2025). These approaches provide methodological context; they are not additional evaluated baselines. The models tested here, their supervision, and their scoring protocols are specified in Evaluation Protocol.

## CORPUS AND ANNOTATION

### Task Definition and Label Schema

Given a Chinese job-ad sentence  $x$ , the task is to predict non-overlapping spans  $(s, e, t)$ . Here,  $s$  and  $e$  are zero-based, end-exclusive Unicode-code-point offsets, and  $t \in \{L, K, S, T\}$ . Each span must reproduce the contiguous substring  $x[s : e]$ , including its spelling and internal spaces. The annotation describes a requirement in the advertisement; it does not measure an applicant’s ability. Figure 1 summarizes how annotation supports model comparison and agreement assessment. The handbook assigns L to language requirements, including language names, proficiency levels, examinations, and certificates. K covers facts, principles, theories, and fields of study. S covers

Chinese-SkillSpan: annotation, model evaluation, and agreement



Human-reference scoring, Silver supervision, and annotator agreement have different targets.

**Figure 1.** Chinese-SkillSpan workflow. Human-reference evaluation, Silver-supervised model fitting, and independent annotation agreement use distinct targets. Figure 2 details the annotation cohorts.

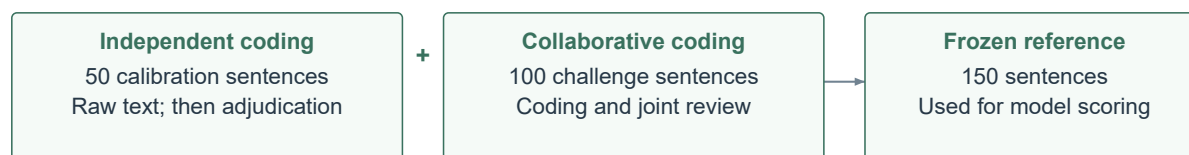
occupational actions, methods, and tool use. T covers transversal capabilities such as communi-  
cation, cooperation, cognition, and self-management. These categories draw on ESCO and on the  
distinction between knowledge and skills in the European Qualifications Framework (European  
Commission, 2024; Council of the European Union, 2017).  
The project K label includes substantive knowledge and recruitment proxies: educational qualifi-  
cations, non-language qualifications, and explicit industry-background requirements. These prox-  
ies are an operational extension, not an assertion that credentials or experience measure knowl-  
edge under ESCO or the EQF. Accordingly, K scores measure extraction of this combined project  
category. The frozen labels do not identify these subtypes, so a knowledge-only evaluation re-  
quires an additional human review. The boundary and context rules below define the operational  
task.  
Table 2 gives the sample sizes and uses of the cohorts in Figure 2.

**Corpus Construction and Annotation Layers**

The corpus contains 22,840 sentences from the AI, Cloud, public-sector, and graduate-recruitment  
collections, with an original split of 17,460 training, 2,143 development, and 3,237 test sentences.  
These are collection labels, not mutually exclusive occupations or acquisition routes. The original  
corpus partition is separate from the smaller labeled experiments below; the 3,237-sentence orig-  
inal test partition is not a human-annotated test set used in this paper. Table 2 distinguishes the  
corpus from the smaller subsets used for annotation, supervision, and evaluation.  
Texts came from three routes: commercial collections purchased from MacroData (MacroData,

## Annotation and review: distinct cohorts and purposes

### A. Historical human-reference construction

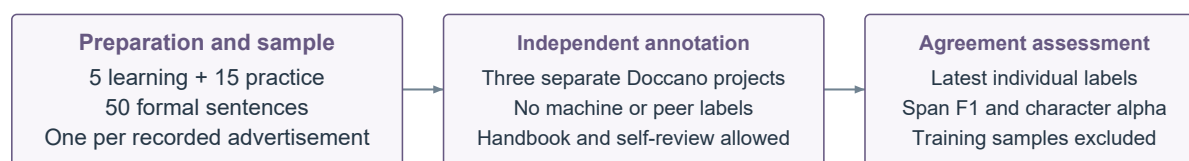


### B. Silver labels and sampled review



Historical coverage: 500 unique human-coded or reviewed sentences

### C. Separate final-handbook agreement study



**Figure 2.** Annotation cohorts and their roles. Historical coding and review cover 500 unique sentences. The separate agreement study uses the latest individual annotations after self-review; its five learning and 15 practice sentences are excluded from agreement.

nd); public-institution recruitment announcements collected by the research team; and the Recruitment Dataset (dataset ID 163746) on Alibaba Cloud Tianchi (Tianchi, nd). For public-institution announcements, team members selected pages from government portals, university and hospital websites, and public recruitment platforms across regions of China. A crawler revisited the registered URLs weekly or monthly. Screening and cleaning produced the public-institution subset. Appendix D records the available source metadata and gaps in the original collection history. We reserve *original Silver* for the initial generated-label pool and *expanded Silver* for the later training pools. Earlier and revised supervision are denoted B1 and B2 in experiment files, respectively. The original annotation study contains 2,601 records: a 150-sentence human reference, named Gold150 in the released files, and 2,451 records with LLM-generated Silver labels, named Silver-plus in archived files. The Silver set includes 100 initially reviewed records, 730 further records from the model-disagreement queue, and 1,621 historical extraction records. The 980-record disagreement queue contributed the reference sentences and the first 830 Silver records. The human reference combines 50 calibration sentences with a non-overlapping 100-sentence challenge sample. The calibration sentences were coded independently from raw text and then adjudicated; the challenge sample was collaboratively coded and reviewed. Both informed guideline development. Their final annotations were fixed for evaluation.

The reference contains 663 spans: 448 S, 125 K, 88 T, and two L. Eight sentences contain no annotated span. Its 150 complete sentence texts are distinct after NFC normalization. The two original 100-sentence review samples are separate from the reference, giving 350 unique sentences with human coding or review at this stage. The historical 15-sentence assisted-review subset is nested within this coverage. Appendices B and D give cohort counts and sampling details.

**Table 2.** Annotation and supervision subsets. Record counts follow the experiment inputs; a record may repeat a sentence. The independent agreement sample is separate from the model reference.

| Layer                          | Records     | Support / composition                                           | Role                                           |
|--------------------------------|-------------|-----------------------------------------------------------------|------------------------------------------------|
| Original corpus                | 22,840      | Four source collections                                         | Source corpus and original partition           |
| Human reference                | 150         | 663 spans; 8 empty; 50 + 100 cohorts                            | Primary diagnostic evaluation                  |
| Original Silver pool           | 2,451       | 100 reviewed; 730 conflict-origin; 1,621 other                  | Candidate labels; version-specific inclusion   |
| Initial review                 | 100         | Versioned review layers (Appendix B)                            | Review of LLM labels; separate from evaluation |
| Post-generation review         | 100         | Machine: 144 spans; reviewed: 148                               | Sampled machine-human quality check            |
| JobBERT train / dev            | 2,156 / 169 | Earlier/revised labels on matched texts                         | Historical supervision and selection           |
| Qwen revised-label train / dev | 2,150 / 169 | 3,646 / 345 spans; 869 / 82 empty                               | Earlier adapter fitting and selection          |
| Additional dual-review QA      | 150         | Both reviewers complete; 101 identical span sets                | Quality check of the new Silver portion        |
| Blinded agreement              | 50          | Three independent coders; five + 15 training sentences excluded | Handbook agreement; individual labels          |
| Expanded Silver (Codex)        | 9,540       | 8,842 / 348 / 350 train / validation / test                     | Training, selection, and Silver-target checks  |

**Expanded supervision and annotation profile.** The adopted Codex pool contains 9,540 sentences from earlier supervision and additional cloud, public-sector, and listed-company collections. Selection prioritized requirement-related keywords. The supplementary 9,646-sentence pool differs in annotation channel and retained records. Appendix D records sampling and exclusions; Appendix C reports the two experiment-specific variants.

Two reviewers checked a further stratified sample of 150 newly annotated sentences, called QA150 in the released summaries. The sample contains 60 cloud, 45 public-sector, and 45 listed-company sentences and excludes the original 350 human-coded or reviewed sentences by identifier and normalized text. Both reviewers completed separate submissions with machine annotations visible. Historical human coding and review thus cover 500 unique sentences; the evaluation reference remains the original 150. The separate 50-sentence blinded agreement study is described below and is not added to this historical coverage count.

The disagreement-selected human reference emphasizes difficult cases and differs from the supervision pool. Empty annotations occur in 40.4% of original Qwen training records and 48.5% of development records, compared with 5.3% of reference sentences (Table 2). Occupational skills also dominate all five profiled layers (59.7–67.6%). Reference spans also have a shorter median length (four code points) than supervision spans (six). These selection and annotation differences preclude treating the reference as a representative sample of recruitment sentences. They motivate type and length diagnostics, but do not by themselves identify the cause of exact-versus-relaxed score gaps. Figure 3 shows length and type distributions; Appendix B records the medians.

## Annotation Guidelines and Adjudication

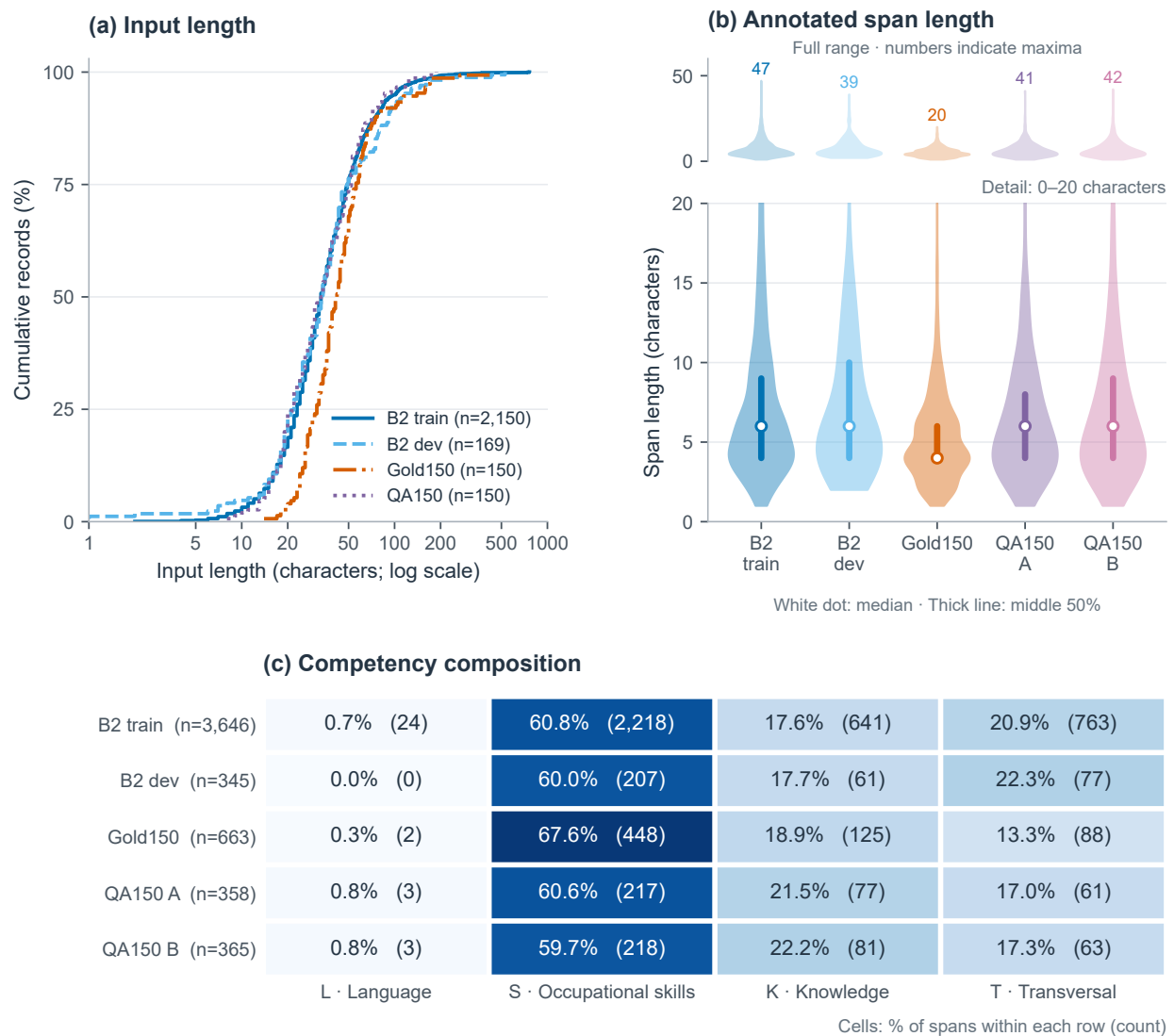
The guidelines address three decisions that a list of skill terms cannot settle: whether a clause is eligible, how much text to retain, and which type fits its context. Annotation begins by determining whether a clause expresses a requirement for the candidate. Benefits, applicant-side procedures, company promotion, and pure outcomes do not qualify on their own; recruitment activities performed as occupational duties are assessed separately. Mixed clauses are considered separately. A sentence with no competency mention receives an empty annotation; insufficient context or an unresolved interpretation is recorded as uncertainty.

We adapted SkillSpan’s semantic-completeness and coordination principles for Chinese annotation (Zhang et al., 2022b). Our handbook asks annotators to select the shortest source span that preserves the complete requirement, without a fixed length cap. Generic proficiency triggers usually remain outside the span, but a modal can remain inside a complete colloquial capability expression. Necessary action heads and complements are retained. Coordinated activities are split only when each remains interpretable; a shared action and its objects stay together (Figure 4). Experience expressions are assessed in three steps. Annotators first distinguish a named occupational activity from generic tenure or industry background. Generic duration alone is excluded; explicit industry-background requirements receive the project K proxy label. For an activity, the annotator then removes the experience qualifier provisionally and checks whether the remaining words still name the required activity in context. If only an event name or incomplete fragment remains, the qualifier is retained. Finally, coordinated items are split only when each passes the same check. An internet-company operations activity is S; a requirement for an internet-industry background is K. Appendix A provides contrasting examples.

Type depends on what the clause requires: using SQL supports S, whereas knowledge of SQL principles supports K. A predicate can determine type without being part of the selected span. Familiarity alone does not decide type. General communication and coordination remain T inside duty clauses, whereas technical operations receive S. Language examinations receive L; educational and non-language qualifications receive K. Figure 4 shows original corpus examples with their archived annotations, and Appendix A gives further examples and rule sources.

Adjudication treats scope, boundaries, and type as separate decisions (Artstein and Poesio, 2008). There is no total priority order among L, K, S, and T. Type uncertainty is recorded separately from accepted labels. The current handbook permits provisional S in a working adjudication record, but requires the entire unresolved record to be excluded from final training targets. This bookkeeping convention does not license assigning S by default in the accepted data. Same-boundary alternatives, nested candidates, and unresolved overlaps are kept in the adjudication record rather than duplicated in the flat annotation layer. Appendix A gives the full rules and their history. The displayed corpus examples retain historical annotations; they are distinguished from





**Figure 3.** Benchmark annotation profile. (a) Empirical cumulative input-length distributions, including records with no spans; the horizontal axis is logarithmic. (b) Span-length violins: the main view enlarges 0–20 characters, covering 96.4–100.0% of spans in each layer; the upper overview retains the full range and labels the maxima. Both views use the same full-data density and equal maximum widths. White dots and thick lines mark medians and interquartile ranges. (c) Within-layer type percentages and span counts. B2 train/dev are the original 2,150/169-record Qwen manifests; Gold150 is the frozen 150-sentence reference. QA150 A/B are the two assisted-review layers of the same additional 150 sentences, shown once in (a). Lengths count source Unicode code points, including punctuation and spaces, and are not directly comparable with English token counts; repeated B2 records are retained. These historical supervision and review profiles exclude the independent three-annotator sample and the full expanded pool.

**L** Language   **K** Knowledge   **S** Occupational skill   **T** Transversal competence

**a) Language and transversal requirements**

1802-s0008 | excerpt

4、良好的英文读写能力<sup>L</sup> 5、优秀的沟通能力<sup>T</sup> 和团队合作精神<sup>T</sup>

Good English reading and writing, communication, and teamwork are distinct requirements.

**b) Knowledge and programming skills**

1806-s0005 | full sentence

2 深刻理解面向对象<sup>K</sup>、熟悉设计模式<sup>K</sup>、多线程和跨平台编程<sup>S</sup>；

Understanding object orientation and design patterns differs from multithreaded and cross-platform programming.

**c) Coordinated occupational actions**

1803-s0002 | full sentence

3、独立操作竞价<sup>S</sup>，如扩展长尾关键字<sup>S</sup>，调整价格<sup>S</sup>，优化曝光与转化率<sup>S</sup>，并协助完成Listing优化<sup>S</sup>；

Bid management includes keyword expansion, price adjustment, exposure/conversion optimization, and listing optimization.

**d) Colloquial and interpersonal capabilities**

1973-s0006 | excerpt

4、会聊天<sup>T</sup>，有服务意识<sup>T</sup>，善于与人沟通<sup>T</sup>，具备较好的协调<sup>T</sup>、沟通能力<sup>T</sup>；

Conversation, service awareness, interpersonal communication, and coordination express transversal capabilities.

**e) Experience requirements**

1896-s0005 | full sentence

2、5年以上售前顾问<sup>S</sup>或项目管理<sup>S</sup>经验，有一定数量的大项目售前<sup>S</sup>经验；

The record requests presales or project-management experience; generic duration and experience qualifiers lie outside the recorded spans.

**Figure 4.** Historical corpus annotations from the frozen human reference, with L/K/S/T spans and source IDs. Panels (a) and (d) are contiguous excerpts; the others show complete stored sentences. Highlights retain the archived v4.2.10 labels. These are the labels used in evaluation, including historical boundary choices. Appendix A explains the current rules and their relationship to these examples. English glosses are display aids.

current-handbook illustrations in Appendix A. Each completed experiment retains its original labels.

### Human Annotation and LLM-Assisted Labeling

Human annotation and adjudication established the reference labels and informed the handbook.

The authors used the official OpenAI Codex interface for adopted Silver annotation; Codex names the interface, while Table 13 identifies recorded model names and roles. Initial sample review preceded bulk annotation. Automated checks verified substrings, offsets, labels, and coverage; human review addressed interpretation.

Bulk annotation used the handbook available at generation time. Later amendments were recorded separately; completed experiments remained unchanged.

Codex supplied labels for the replaced subset in the adopted expanded variant; the alternative variant is supplementary. Earlier supervision components remain in both pools. Unresolved records were excluded. Appendices A and D document versions and inclusion decisions.

### Annotation Quality and Reference Freezing

Independent coding assesses handbook use without machine or peer suggestions; assisted review evaluates proposed labels. We report final-handbook agreement from the separate three-annotator blinded study (Table 3). Historical coding and review used different samples and handbook versions (Appendix B); their scores cannot track independent reliability over time.

**Blinded coding protocol.** Three annotators used handbook B.sop\_v4.2.14 after familiarization and training on sets of five and 15 sentences, following the training-and-feedback approach discussed by Klie et al. (2024). They then independently coded the same sample of 50 sentences drawn from existing Silver supervision in separate Doccano projects, blinded to both machine suggestions and one another's labels. Training sentences are excluded from formal agreement. The sample contains one sentence from each of 50 recorded advertisement identifiers and assesses handbook agreement. It was not reserved from expanded-model training. Appendix B gives source counts and sampling limits; Appendix D reports intersections with model inputs. Annotators could consult the handbook and revise their own labels during individual self-review. We computed agreement from their latest separate exports, retaining all 50 sentences and all three coder pairs. These individual layers were not adjudicated into model-test labels. Typed exact F1 measures agreement on complete spans without designating a coder as gold (Hripcsak and Rothschild, 2005); boundary-only F1 ignores type. Nominal Krippendorff's  $\alpha$  uses L/K/S/T/O assignments to all 2,111 source code points. Percentile 95% intervals use 10,000 source-stratified sentence-bootstrap samples, retaining all three annotations together. These intervals condition on the observed source mixture and coders.

Mean typed F1 is 0.818 (95% CI [0.732, 0.894]). Boundary F1 is higher at 0.850 (95% CI [0.766, 0.923]), indicating additional type disagreement. Character-level  $\alpha = 0.907$  (95% CI [0.851, 0.954]) measures agreement at a different unit from complete spans (Artstein and Poesio, 2008). Of the 34 fully agreed sentences, 11 are empty. The two L spans occur in one sentence, limiting category-level conclusions. The reproduction materials include the latest individual labels and diagnostics for this independent coding study.

**Table 3.** Independent blinded agreement among three annotators on 50 sentences after individual self-review. Character coefficients are Cohen’s  $\kappa$  unless marked  $\alpha$ . Brackets give 95% bootstrap intervals for typed exact F1. “Equal” counts identical sentence-level span sets. Training samples are excluded.

| Comparison                                       | $n$ | Character      | Typed F1                | Details                                       |
|--------------------------------------------------|-----|----------------|-------------------------|-----------------------------------------------|
| <i>Final-handbook independent blinded coding</i> |     |                |                         |                                               |
| A–B                                              | 50  | 0.880          | 0.793<br>[0.694, 0.884] | Boundary F1 0.828; equal 36/50                |
| A–C                                              | 50  | 0.930          | 0.848<br>[0.752, 0.926] | Boundary F1 0.876; equal 39/50                |
| B–C                                              | 50  | 0.909          | 0.814<br>[0.721, 0.899] | Boundary F1 0.847; equal 37/50                |
| Mean / three-rater                               | 50  | $\alpha$ 0.907 | 0.818<br>[0.732, 0.894] | Mean boundary F1 0.850; all three equal 34/50 |

## Ethics and Data Use

The study analyzed recruitment advertisements and did not recruit job applicants or collect applicant responses. Annotation and review were performed by research-team members. Institutional ethics approval was not sought; no formal exemption is claimed. The authors confirm permission to use the source materials for academic research. Permissions to conduct research and rights to redistribute third-party recruitment text are recorded separately in the component-specific access terms. Privacy screening of the released text is reported in Appendix D.

## EVALUATION PROTOCOL

The evaluation compares prompted extraction, measures gains from Silver-supervised Qwen adaptation, and tests encoder initialization under shared task-specific settings. Type, span-length, and empty-sentence diagnostics complement aggregate scores. Primary Chinese comparisons use the human-finalized reference. Supplementary Silver-validation and Silver-test scores measure agreement with generated labels and remain separate from human-reference results.

## Datasets and Splits

The original corpus was split by source document: training, development, and test contain 1,600, 200, and 200 document identifiers, respectively, with no identifier shared across partitions. The subsequent B2 supervision and Gold150 reference were assembled separately from the original test partition. We therefore distinguish the original corpus split from these later labeled experimental splits.

The frozen B2 training package records sentence-level exclusion checks. JobBERT uses 2,156 training and 169 development records. Removing six training records that match three development texts after NFC normalization gives Qwen’s 2,150/169 split. In this Qwen and encoder split, training and development share no exact or NFC-normalized complete texts, and neither contains a Gold150 sentence by ID or normalized text. These checks confirm that reference sentence texts are absent from the training and development files.

A complementary audit groups sentences by the original stored source-document identifiers. B2 training and development share one group, and training and Gold150 share 59 groups, containing 143 reference sentences. These are different sentences from shared source documents, not duplicate reference sentences in training. A document can be a multi-post recruitment announcement. Thus the later experiments support held-out-sentence comparisons, while source-document generalization remains a separate evaluation question. Appendix D reports repeated training records, normalization checks, and expanded-pool group counts. Supplementary expansion experiments use 8,842/348/350 training, validation, and test sentences from the adopted Codex pool, and 8,943/351/352 from the alternative pool. Both combine earlier supervision with additional annotations, but differ in annotation channel and the examples retained after checking. Records retained after the channel replacement keep their split assignments. For Qwen, Silver-validation exact F1 selects the checkpoint, Silver-test F1 measures agreement with held-out generated labels, and the human reference provides an additional evaluation. Reference sentences are excluded from gradients and checkpoint selection under the recorded sentence-level checks. Variants and results are in Appendix C; split checks are in Appendix D.

### Evaluation Metrics

To distinguish exact extraction from approximate boundary agreement, the primary metric is typed exact-span micro-F1: a prediction matches only when both character boundaries and the LSKT type agree with the reference. Typed relaxed micro-F1 uses one-to-one greedy matching of same-type spans with intersection-over-union (IoU) of at least 0.5:

$$\text{IoU}(u, v) = \frac{|u \cap v|}{|u \cup v|}, \quad \text{IoU}(u, v) \geq 0.5. \quad (1)$$

Here,  $u$  and  $v$  are the sets of character positions covered by a predicted and a reference span, and  $|\cdot|$  counts positions. For either matching rule, TP counts matched pairs, FP unmatched predictions, and FN unmatched reference spans, summed across sentences:

$$P_{\text{micro}} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad R_{\text{micro}} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad F_{1,\text{micro}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}. \quad (2)$$

All 150 human-reference sentences are scored. Expanded Qwen experiments also report scores on their Silver validation and test sets. Partially rejected responses retain accepted spans; malformed responses or responses with no accepted spans are scored as empty predictions. Empty-empty sentences add no span counts. Appendix C summarizes parser decisions and matching order and links the full scorer specification. Boundary exact micro-F1 matches span boundaries after collapsing types. Per-type scores retain the original L/K/S/T labels and include all reference sentences, including sentences without spans of the evaluated type.

Across three training seeds, we report the mean and sample standard deviation:

$$\bar{f} = \frac{1}{n} \sum_{i=1}^n f_i, \quad s_f = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (f_i - \bar{f})^2}, \quad n = 3. \quad (3)$$

Inference-only and expanded-pool results are single runs unless otherwise stated. Appendix C documents resampling procedures.

## Models and Baselines

JobBERT-zh provides a sequence-labeling baseline with a Chinese recruitment-domain-pretrained encoder and a CRF head. We compare Qwen2.5-14B-Instruct (Qwen) (Qwen Team, 2025) with and without a project LoRA adapter (Hu et al., 2022) for generative extraction. The shared-guideline inference comparison includes GPT (API), Claude (API; Sonnet 4.5), DeepSeek-V4-Pro (DeepSeek), Kimi k2.6 (Kimi), and Qwen. DeepSeek is evaluated with thinking enabled and disabled, giving six configurations. The adopted Silver annotation used the recorded Codex selection, whereas the GPT baseline used a separate API configuration (Table 13). Their errors need not be independent. Shared guidelines and historical assisted review can reinforce annotation conventions; agreement with these conventions does not independently validate the teacher. Appendix C records service identifiers, access routes, and inference settings.

## Implementation Details

**Supervision and model selection.** JobBERT conditions start from the same released encoder and CRF. Earlier supervision (B1 in the experiment files) uses the historical extraction labels; revised supervision (B2) combines updated guidance, LLM annotation, and adjudication. The conditions share training and development texts, with label changes in 1,440 training records and 107 development records. Development typed exact F1 selects checkpoints, so this comparison changes training supervision and selection targets together. Qwen starts from its pretrained Instruct weights and fits a fresh LoRA adapter for each of seeds 42, 43, and 44 on the revised 2,150/169 split. Training uses a fixed system instruction, user template, and five schematic examples. Targets contain the literal span text, occurrence index, and type. Loss applies to the assistant JSON and closing turn token. Training uses BF16, and development exact F1 selects the adapter. Table 4 gives the remaining settings. Historical JobBERT implementation gaps are documented in Appendix D.

**Table 4.** Original JobBERT and Qwen training settings. Both select checkpoints by development typed exact F1; reproduction notes retain implementation and selection records.

| Setting              | JobBERT-zh B1/B2     | Qwen LoRA                        |
|----------------------|----------------------|----------------------------------|
| Learning rate        | $2 \times 10^{-5}$   | $10^{-4}$                        |
| Batch size           | 16                   | 1; 16-step gradient accumulation |
| Maximum input length | 256 tokens           | 8,192 tokens                     |
| Training limit       | 6 epochs; patience 2 | 6 epochs; patience 2             |

Qwen uses LoRA with rank 16, scaling 32, and dropout 0.05, applied to the attention q/k/v/o projections.

**Inference.** Each request contains one sentence’s identifier and text, together with the shared instruction and five fixed examples. A parser converts occurrence-based output into character spans. Qwen baseline and adapter inference use the same base checkpoint, prompts, parser, BF16 precision, deterministic decoding, and 2,048-token output cap. DeepSeek uses caps of 2,048 tokens without thinking and 8,192 with thinking. Appendix C lists service settings and output checks.

## Chinese-Supervised Encoder Comparison

To examine whether taxonomy-informed initialization helps this Chinese task, we compare XLM-R-large (Conneau et al., 2020) and ESCOXML-R (Zhang et al., 2023). Both receive fresh nine-label BIO heads and are fine-tuned on the same B2 split of 2,150 training and 169 development records. They use AdamW, learning rate  $2 \times 10^{-5}$ , batch size 8, at most 20 epochs, and seeds 42, 43, and 44. Development typed exact F1 selects each checkpoint.

Both models process the original Chinese characters. The BIO head labels the first subword of each character; predictions from chunks of at most 128 characters are joined, including the remainder of long sentences. Tokenizer placeholders preserve positions for whitespace and otherwise discarded characters. Scoring uses the original text and offsets. The reference is excluded from fitting and checkpoint selection in these runs.

We report typed exact and relaxed F1, boundary-only F1, and both four-type and K/S/T macro-F1. The latter sensitivity summary removes the disproportionate macro weight of the two L spans without changing the four-type prediction task. Training heads and budgets differ from JobBERT and Qwen, so cross-architecture scores remain descriptive. Paired uncertainty is calculated with the same source-group bootstrap used for the main comparisons (Appendix C).

## RESULTS AND ANALYSIS

### Inference on the Human Reference

GPT (API) achieved the highest typed exact micro-F1 among the six configurations, followed by DeepSeek with thinking disabled (Table 5). Their paired difference is 0.020, with a 95% source-group bootstrap interval of  $[-0.027, 0.061]$  (Table 6). The interval includes zero, so the point scores do not establish a stable ordering. These are comparisons of the recorded configurations, which use shared instructions but different inference budgets.

**Table 5.** Shared-guideline inference configurations on the human reference ( $n = 150$ ). P/R are typed exact precision/recall. Each configuration is a single run. Inference budgets are configuration-specific. Qwen uses no project adapter.

| Recorded configuration     | P     | R     | Exact F1 | Relaxed F1 |
|----------------------------|-------|-------|----------|------------|
| GPT (API)                  | 0.676 | 0.658 | 0.667    | 0.766      |
| DeepSeek<br>(non-thinking) | 0.651 | 0.643 | 0.647    | 0.761      |
| Claude (API)               | 0.598 | 0.597 | 0.598    | 0.709      |
| DeepSeek<br>(thinking)     | 0.713 | 0.505 | 0.591    | 0.671      |
| Kimi                       | 0.581 | 0.563 | 0.572    | 0.714      |
| Qwen                       | 0.507 | 0.281 | 0.361    | 0.470      |

Both DeepSeek rows use the same model. Non-thinking and thinking denote inference modes, with output caps of 2,048 and 8,192 tokens respectively. Table 13 lists the recorded identifiers, access routes, and settings. Cohort scores and detailed output-status counts accompany the reproduction materials.

**Table 6.** Paired typed exact-F1 differences on the human reference, with 95% intervals from resampling the 61 source-ID groups.

| Comparison                            | Difference | 95% interval    |
|---------------------------------------|------------|-----------------|
| GPT (API) minus DeepSeek non-thinking | +0.020     | [−0.027, 0.061] |
| Qwen LoRA 42 minus no adapter         | +0.138     | [0.069, 0.200]  |
| Qwen LoRA 43 minus no adapter         | +0.202     | [0.133, 0.269]  |
| Qwen LoRA 44 minus no adapter         | +0.197     | [0.123, 0.268]  |
| Qwen LoRA seed mean minus no adapter  | +0.179     | [0.111, 0.242]  |
| ESCOXLM-R minus XLM-R-large           | +0.016     | [−0.005, 0.035] |

Each bootstrap draw resamples complete source-ID groups with replacement and uses the same draw for all models. Intervals condition on the recorded runs and reference; they are pointwise and do not establish independence from guideline development.

### Fine-Tuning with Silver Labels

All three Qwen adapters improved exact precision and recall over the no-adapter baseline under the same inference protocol (Table 7, panel A). Mean typed exact micro-F1 increased from 0.361 to  $0.540 \pm 0.035$ . Each seed’s paired interval against the baseline also excludes zero (Table 6), conditional on the recorded runs and reference sentences. Partially rejected responses fall from 69 without an adapter to 2, 4, and 1 after adaptation. Valid spans from partially rejected responses remain in scoring; partial rejection differs from a whole-response failure. The measured benefit is end-to-end extraction, combining content prediction and valid output generation. A format-controlled intervention would be needed to separate those effects.

**Table 7.** Evaluation of Silver-supervised fine-tuning on Gold150 ( $n = 150$ ). F1 scores (0–1) are mean  $\pm$  sample SD over seeds 42, 43, and 44; Qwen without an adapter is a single run.

| A. Qwen: fixed inference protocol                   |             |                   |                   |                   |
|-----------------------------------------------------|-------------|-------------------|-------------------|-------------------|
| Condition                                           | Train / dev | Typed exact       | Typed relaxed     | Boundary exact    |
| No adapter                                          | —           | 0.361             | 0.470             | 0.414             |
| B2 LoRA                                             | 2,150 / 169 | $0.540 \pm 0.035$ | $0.662 \pm 0.033$ | $0.571 \pm 0.038$ |
| B. JobBERT-zh: historical supervision and selection |             |                   |                   |                   |
| Supervision and selection labels                    | Train / dev | Typed exact       |                   |                   |
| Earlier (B1)                                        | 2,156 / 169 | $0.142 \pm 0.014$ |                   |                   |
| Revised (B2)                                        | 2,156 / 169 | $0.554 \pm 0.005$ |                   |                   |

A: same base checkpoint, frozen messages, occurrence schema, parser, and decoding; adapters selected by B2 development F1. Boundary exact matches spans with types collapsed. B: same initial encoder/CRF and matched training/development texts, with both label sets changed. Limitations and Responsible Use covers split and checkpoint limitations; the per-seed results gives per-seed Qwen scores for all three training runs.

The historical JobBERT configuration achieved higher reference agreement with revised supervision (Table 7, panel B). The two conditions start from the same encoder and CRF and use matched training and development texts, but revise both sets of labels. Their difference therefore reflects changes to supervision and checkpoint selection together.



## Supplementary Analyses

The Qwen improvements extend across the three reported span-length bins: every adapter increases exact recall in each bin (Figure 5). All three adapters produced fewer false-positive spans on the eight empty-reference sentences. This describes those eight examples and does not establish a population false-positive rate. The figure shows individual-seed results; Table 8, panel B, compares aggregated category scores across models.

Exploratory experiments with expanded supervision appear in Appendix C. Those runs change the source composition, annotation targets, and retained examples together. They do not isolate an expansion effect. Available diagnostics do not explain the lower expanded JobBERT scores.

## Chinese Encoder and Type Comparisons

The Chinese comparison does not establish a reliable advantage of ESCO initialization. The encoders achieved mean exact F1 of 0.522 (XLM-R-large) and 0.538 (ESCOXLM-R), with all six prediction files independently rescored against the original reference (Table 8, panel A). The mean difference was 0.016; its paired 95% interval was  $[-0.005, 0.035]$ . Four-type macro-F1 is 0.607 and 0.622; excluding the two-span L class gives descriptive K/S/T macro values of 0.542 and 0.562, respectively. This check does not replace the four-type metric.

Silver-supervised Qwen adaptation improved both S and K extraction: mean typed exact F1 increased by 0.148 and 0.210, respectively (Table 8, panel B). Each adapter improved K, S, and T relative to the no-adapter run (Figure 5). Qwen has the highest S mean and ESCOXLM-R the highest K mean among these three adapted models. These cross-model comparisons are descriptive because prediction heads, training budgets, and inference procedures differ.

**Table 8.** Chinese model diagnostics on Gold150 (0–1). Results across three training seeds are mean  $\pm$  sample SD; Qwen without an adapter is a single run.

| A. Chinese encoders: overall F1      |                   |                   |                   |                   |                   |
|--------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Encoder                              | Typed exact       | Typed relaxed     | Boundary exact    | Macro L/K/S/T     | Macro K/S/T       |
| XLM-R-large                          | $0.522 \pm 0.022$ | $0.650 \pm 0.021$ | $0.571 \pm 0.022$ | $0.607 \pm 0.016$ | $0.542 \pm 0.021$ |
| ESCOXLM-R                            | $0.538 \pm 0.021$ | $0.662 \pm 0.022$ | $0.582 \pm 0.024$ | $0.622 \pm 0.010$ | $0.562 \pm 0.013$ |
| B. Typed exact F1 by competency type |                   |                   |                   |                   |                   |
| Model                                | S ( $n = 448$ )   | K ( $n = 125$ )   | T ( $n = 88$ )    | L ( $n = 2$ )     |                   |
| Qwen, no adapter                     | 0.377             | 0.302             | 0.380             | 0.000             |                   |
| Qwen B2 LoRA                         | $0.525 \pm 0.039$ | $0.512 \pm 0.063$ | $0.638 \pm 0.017$ | 0.800             |                   |
| XLM-R-large                          | $0.506 \pm 0.024$ | $0.495 \pm 0.037$ | $0.627 \pm 0.006$ | 0.800             |                   |
| ESCOXLM-R                            | $0.518 \pm 0.029$ | $0.537 \pm 0.009$ | $0.631 \pm 0.020$ | 0.800             |                   |

All adapted models use the B2 split (2,150/169 training/development records) and seeds 42/43/44. Encoders use fresh linear LSKT heads; Qwen uses a LoRA adapter and the fixed inference protocol in Table 7. All four types are predicted jointly. Boundary exact collapses types; macro summaries equally weight their named categories. For L, each trained run has 2 TP, 1 FP, and 0 FN; Qwen without an adapter has 0 TP, 1 FP, and 2 FN. L seed SD is omitted because two reference spans cannot establish reliability.

## Boundary and Type Disagreements in Human Coding

The three-annotator study shows that character agreement alone understates the difficulty of reproducing complete typed spans (Table 3). Boundary-only agreement exceeds typed agreement, so disagreements concern both extent and type. Historical calibration cases motivated the distinction between job-use tools and knowledge, and between general communication and technical operations. The final-handbook sample differs from Gold150 in texts, label version, and annotation procedure. Its human–human F1 of 0.818 therefore cannot be interpreted as a ceiling for model–reference F1 on Gold150. A common independently annotated sample would be needed for that comparison.

## Implications for Benchmark Use

The no-adaptor Qwen score rises from 0.361 under exact matching to 0.470 under overlap-tolerant matching (Table 5). This difference shows how boundary tolerance affects the evaluation. The prediction–reference examples in Appendix C illustrate boundary and type mismatches, but have not been adjudicated into a linguistic error taxonomy.

Agreement with generated labels does not by itself establish agreement with human annotations. The Qwen run using the adopted expanded Codex pool achieves 0.809 on its Silver test and 0.585 on the human reference (Table 14). The two sets differ in both texts and targets, so the gap cannot estimate Silver-label error, but it illustrates why both evaluations are needed.

The prompted GPT configuration remains above all reported adapted models on Gold150, including expanded-pool Qwen. We therefore claim a benefit of Qwen adaptation relative to its own base checkpoint, not superiority over the prompted configurations. Matched Qwen runs provide the clearest adaptation evidence because inference is fixed. JobBERT and expansion studies jointly change labels, selection, or source composition, so they cannot isolate those effects. Type, length, and empty-sentence diagnostics qualify aggregate scores.

## LIMITATIONS AND RESPONSIBLE USE

**Reference design and label scope.** Gold150 is a disagreement-selected reference used in handbook development. The separate 50-sentence agreement study tests use of the final handbook; it does not supply a new held-out model test. Human annotation agreement and model scores use different samples and targets. K includes qualification and industry-background proxies, and L has only two reference spans. Independent annotation of new, advertisement-isolated texts and human review of K subtypes remain necessary to establish broader validity.

**Historical supervision and access.** The reference, original supervision, and agreement study use different handbook versions. Their labels remain frozen, so final-handbook consistency does not establish compatibility of all historical targets. The original corpus has disjoint source-document partitions, and later B2 checks exclude reference sentences from supervision. The later experimental splits nevertheless share source-document groups, which limits claims about new-document generalization. Incomplete pretraining records leave additional exposure uncertain. Public labels and predictions support inspection and rescoring, while complete expanded-pool inputs are not publicly available for retraining. Reuse must respect the component-specific source permissions.

**Experimental scope.** The Qwen comparison measures end-to-end gains from adaptation, including improved output validity. It does not isolate a semantic-extraction effect. Prompted configurations differ in inference budgets and access routes, expanded-pool studies use single seeds, and

three-seed SD is imprecisely estimated. Source-group bootstrap intervals address dependence among recorded reference prefixes but remain conditional on the selected examples and trained models. Eight empty-reference sentences support descriptive error counts only.

## CONCLUSION

Chinese-SkillSpan provides a benchmark for delimiting and typing competency mentions in Chinese job advertisements. Its contributions are an original-text resource with distinct human and Silver labels, an ESCO-informed operational handbook for Chinese boundaries and contextual types, and annotation-quality evidence that separates independent agreement from assisted review. Fixed character offsets, released predictions, and a shared scorer make model errors inspectable. Three independent annotators achieved mean typed exact agreement of 0.818 after individual self-review under the final handbook. Qwen adaptation improved end-to-end extraction against its own base checkpoint, while the prompted GPT configuration remained strongest by point score. The Chinese encoder comparison did not establish an ESCO initialization advantage. Together, these experiments demonstrate use of the resource for diagnostic comparison and identify the additional annotation and independent testing needed for broader deployment.

## DATA AVAILABILITY

The dataset archive (v0.1.3; DOI: 10.5281/zenodo.22698504) contains the corpus, handbook, 150-sentence human reference, and Silver-plus B2 training/development files; 10.5281/zenodo.22288337 is the concept DOI. The historical annotation-layer release separately provides pseudonymized coding and QA150 exports with original text and offsets. The independent blinded three-annotator study provides frozen labels, source identifiers, agreement code, and results. These later annotation studies are outside the original Zenodo archive.

The Qwen reproduction archive (DOI: 10.5281/zenodo.22851581) contains three B2 LoRA adapters, selected expanded-Silver checkpoints, predictions, configurations, and scoring tools. Complete texts and labels for the adopted 9,540-record Codex pool and the supplementary 9,646-record alternative remain unavailable for public retraining. The six Chinese-supervised XLM-R/ESCOXLM-R checkpoints are linked in the Hugging Face model index; their code, configurations, and predictions are in the audit archive (DOI: 10.5281/zenodo.22937245). The reproduction documentation separates local rescoring of archived predictions from server-reported checkpoint inference.

The JobBERT-zh initialization supplies the domain-adapted encoder and inherited CRF; the B2 continuation supplies continued checkpoints, with seed 42 as its default. The project repository and reproduction guide link code, preprocessing, scoring, and experiment records. Table 17 summarizes access and verification. External-task protocols and results are retained in the supplementary experiment archive.

Reuse terms vary by component. Explicitly covered software and documentation retain Apache-2.0. Author-controlled annotations and research materials permit academic research and peer review with attribution; this permission excludes third-party recruitment text. Model checkpoints and adapters follow their accompanying notices and base-model licences. The data-access guide specifies these terms; Appendix D records earlier versions and access limits.

## APPENDIX A. CODING GUIDE AND EXAMPLES

This guide follows Handbook v4.2.14 and complements Figure 4. The reader-facing v4.2.14 edition clarifies examples for interpretation of the current rules; unresolved cases retain that status. This documentation does not replace the studies' frozen handbooks or labels. The source package contains the bilingual manuals and crosswalk; Section A.4 records version boundaries.

### A.1 Label definitions and context tests

Square brackets mark selected source substrings, with a separate type code after each span. Brackets, colours, and type codes are display aids and are not source characters; boundaries remain identifiable in grayscale. Examples illustrate the handbook, not an additional evaluation sample. *Source key.* E: ESCO's skills hierarchy and transversal concepts (European Commission, 2024); Q: EQF definitions (Council of the European Union, 2017); Z: related annotation guidance in SkillSpan, Appendix B (Zhang et al., 2022b); I: unitization and categorization in agreement analysis (Artstein and Poesio, 2008); P: our V4.2.14 operational decision. Combined marks indicate adaptation, not verbatim rules; Chinese examples and exact boundaries are project illustrations.

**Table 9.** Current LSKT decisions: definitions, examples, and contextual contrasts.

| Label                     | Operational definition                                                                                                                                      | Chinese example                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| L: Language               | Natural-language proficiency, language names, levels, examinations, and certificates. <sup>[E,P]</sup>                                                      | [英语六级] [L]; [日语 N2] [L]                         |
| K: Knowledge              | Facts, principles, theory, and fields of knowledge. <sup>[Q]</sup> Education and non-language qualifications map to K by project convention. <sup>[P]</sup> | [计算机专业] [K];<br>[本科及以上学历] [K]                   |
| S: Occupational skill     | Application of a tool or method, or an occupational activity indicated by the context. <sup>[Q,P]</sup>                                                     | 使用 [SQL] [S];<br>[数据分析] [S]                     |
| T: Transversal competency | General communication, cooperation, cognition, self-management, and dispositions. Duties may also express T. <sup>[E,P]</sup>                               | [客户汇报] [T];<br>[责任心] [T]                        |
| Context test              | Use versus principles distinguishes S from K; general coordination differs from a technical operation. No fixed label priority applies. <sup>[Q,E,P]</sup>  | 了解 [SQL 原理] [K];<br>[协调外部资源] [T];<br>[接口对接] [S] |

### A.2 General annotation decisions

G1–G15 are quick-reference identifiers. Their source sections and amendment IDs (R08–R23) are mapped in the release; they do not replace the handbook's rule numbering.

**Table 10.** Annotation principles for target identity and complete source boundaries.

| Rule | Annotation principle                                                                                                                                                                                | Chinese example                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| G1   | The annotation scheme distinguishes L, K, S, and T. Collapsed categories are used only in secondary evaluation. <sup>[E,P]</sup>                                                                    | [英语六级] [L];<br>[计算机专业] [K];<br>[数据分析] [S]; [责任心] [T]                                  |
| G2   | Final annotations contain non-overlapping spans. A sentence may contain several independent mentions, with each character assigned to at most one span. <sup>[P]</sup>                              | 熟悉使用 [Word] [S], [Excel] [S],<br>[PPT] [S] 等办公软件<br>No additional covering list span. |
| G3   | Each mention is a contiguous source substring that preserves spelling and internal spaces. Any normalization is stored separately. <sup>[Z,P]</sup>                                                 | 熟悉 [Jekins] [S]<br>Keep the source spelling, not Jenkins.                             |
| G4   | Annotators select the shortest span that expresses a complete requirement. Words and identifiers remain intact; completeness takes priority over length, with no fixed length cap. <sup>[Z,P]</sup> | [支持服务] [S]<br>Not the fragment 支持服.                                                   |
| G5   | Generic proficiency triggers usually remain outside the span, while heads needed to express the requirement remain inside. Excluded triggers can still inform type assignment. <sup>[Z,P]</sup>     | 会使用 [Excel] [S];<br>[会聊天] [T];<br>有 [服务意识] [T]                                        |
| G6   | Coordinated items are separated when each remains meaningful in context. Shared qualifiers may stay outside the spans, and implied words are not added. <sup>[Z,P]</sup>                            | 熟悉使用 [Word] [S], [Excel] [S],<br>[PPT] [S] 等办公软件<br>Shared-experience case: see A.3.  |
| G7   | A necessary action, object, or shared head remains within the span when splitting would change its meaning. Span length alone does not determine type. <sup>[Z,P]</sup>                             | [优化曝光与转化率] [S]<br>Isolated metric names would lose the optimization activity.         |

**Table 11.** Annotation principles for scope, experience, type, and unresolved records.

| Rule | Annotation principle                                                                                                                                                                                                                                                                  | Chinese example                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| G8   | Benefits, company promotion, administrative instructions, experience durations, and pure outcomes are outside competency scope on their own. Mixed clauses are assessed separately. <sup>[Z,P]</sup>                                                                                  | 五险一金, 带薪年假 → [];<br>[分析推广效果] [S], 保证流量/用户增长                                                                           |
| G9   | Experience wording is omitted only when the remaining activity preserves the requirement. Necessary practical-experience wording is retained. Explicit industry background follows the handbook's broad-K convention. Experience wording is never added to the source. <sup>[P]</sup> | [机器人竞赛经验] [S]<br>Retain the suffix if a bare competition name changes the requirement; do not label generic 经验 alone. |

Table 11 (continued)

| Rule | Annotation principle                                                                                                                                                                                                                                    | Chinese example                                                                                                                    |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| G10  | Education qualifications include their threshold wording. Language examinations and certificates receive L; non-language qualifications receive K. [Z,P]                                                                                                | [本科及以上学历] [K];<br>[大学英语 6 级] [L];<br>了解 [ISO 27001] [K]<br>Standard familiarity, not a claim of personal certification.            |
| G11  | Tool and method mentions are typed from their predicate context. Occupational use supports S; principles or theory support a complete K phrase. Familiarity alone does not determine type. [Q,P]                                                        | 使用 [SQL] [S];<br>了解 [SQL 原理] [K]                                                                                                   |
| G12  | Occupational functions are distinguished from general cognition, communication, and coordination. A duty clause may contain T; usefulness across industries alone does not establish T. [E,P]                                                           | [编码能力] [S];<br>[沟通能力] [T];<br>[协调外部资源] [T]                                                                                         |
| G13  | Scope, boundaries, and type are resolved separately, without a total label priority. Once scope and boundaries are valid, type-only uncertainty may receive provisional S with alternatives logged; this is excluded from final training targets. [I,P] | Heading alone: 能力提升<br>Insufficient scope evidence: record as unresolved, not a confirmed empty negative.                          |
| G14  | Offsets count original Unicode code points, beginning at zero and excluding the end position. The indexed substring must reproduce the mention exactly; segmentation is a derived view. It does not replace the fixed human-reference offsets. [P]      | text[start:end] == span<br>Keep the original text, internal spaces, and offsets consistent.                                        |
| G15  | Alternatives and overlaps are recorded separately from the final flat layer. An unresolved record is excluded in full from training, with confirmed empty, missing, and unresolved annotations distinguished. [P]                                       | 五险一金 → confirmed [];<br>能力提升 without context → unresolved.<br>Known partial spans do not make an unresolved record training-ready. |

**Interpreting coordinated capabilities.** G6–G7 test whether each selected phrase expresses a requirement in context. They do not require every item to repeat an implied capability noun. For example, 协调 (*coordination*) can express a capability independently, whereas 曝光 (*exposure*) alone does not express the optimization action in 优化曝光与转化率 (*optimize exposure and conversion rate*). Figure 4 shows the recorded historical boundaries; it is not a relabeling under the final handbook. Any unresolved historical case requires a new human decision before its reference span is changed.

### A.3 Shared experience: split only when meaning survives

The V4.2.14 amendment (R23) clarifies a shared-experience construction. [P] Each item remains identifiable in context; exclude the shared experience qualifier and do not add a covering span. The words for internet-company operations denote an occupational activity in this sentence; a

bare requirement for experience in the internet industry would instead use the project K proxy rule. The presence of an industry name alone does not determine type.

4、三年以上大中型 [互联网公司经营] [S] 或 [BI] [S] 相关工作经验, [互联网公司广告运营分析] [S] 优先;

*Three or more years of operations or BI experience in large or medium-sized internet companies; internet-company advertising operations analysis preferred.*

Selected offsets: [9, 16), [17, 19), and [26, 37). BI is S in this context, not by a universal dictionary rule. In contrast, G7 retains the shared action in 优化曝光与转化率 (*optimize exposure and conversion rate*); splitting must not leave bare outcomes. The earlier long span in the initial reviewed LLM-annotation dataset remains historical rather than being silently overwritten.

#### A.4 Version boundaries

The human-reference metadata identify v4.2.10, while the original bulk-generation prompt follows v4.2.12. The current handbook is v4.2.14. Later amendments do not retrospectively change completed annotations. Inspection confirmed v4.2.10 metadata on all 150 reference records; the B2 rows do not carry individual handbook-version fields. One documented difference is present in B2 training record 1838-s0008: its historical two-span annotation retains a shared experience phrase that R23 later splits into three spans. This named record is absent from Gold150 and B2 development. A rule-family screen covers all 2,469 reference, training, and development records, but flags candidates for contextual review rather than proving semantic compatibility. The original targets and scores remain unchanged. Figure 4 reproduces frozen v4.2.10 human-reference annotations, whereas Tables 9–11 illustrate decisions under the current v4.2.14 handbook. Later rule amendments do not retrospectively alter the historical annotations or reported scores. The label-version documentation specifies the handbook and annotation files used in each experiment. The accompanying manuals preserve original and revised spans and their rationale. Human-reviewed Silver-plus remains supervision unless separately designated and isolated for evaluation. Evaluation uses the frozen reference, shared protocol, occurrence schema, parser, and scorer; the reproduction guide maps these terms to their released files (Appendix D).

## APPENDIX B. ANNOTATION-QUALITY EVIDENCE

Historical coding and assisted-review comparisons below use designs distinct from the independent blinded study in Table 3. The initial 50-sentence calibration study involved independent raw-text coding before adjudication under the historical handbook. The sentences were sampled from the 980-record conflict queue. Available exports reproduce 98 exact matches between 211 and 230 spans, giving F1 0.444. Character-level  $\kappa$  is 0.567 for L/K/S/T plus O, over all 2,571 source code points, including spaces and punctuation. All 50 identifiers belong to the calibration portion of Gold150 and none occurs in the B2 training or development identifiers. Thus, the study supplies independent-coding evidence, but is not an additional test cohort. These calculations use the available later exports; exact correspondence with the original frozen files has not been verified. The agreement calculations preserves the arithmetic and supporting counts. The reference's challenge cohort contributes 415 spans (290 S, 83 K, 41 T, one L), and the calibration cohort contributes 248 (158 S, 42 K, 47 T, one L). Seven challenge sentences and one calibration sentence have no annotated span. The identifiers contain 61 source-document prefixes, verified against the original corpus identifiers. The adjudicated reference remains separate from

the original coding and assisted-review layers. A retrospective comparison of the two available individual calibration exports with their adjudicated 50-sentence reference gives typed exact F1 of 0.640 and 0.606 (boundary F1 0.766 and 0.684). These annotators contributed to that reference; the comparison describes adjudication changes and is not independent validation or a human performance ceiling.

Character-level Cohen's  $\kappa$  adjusts observed agreement for agreement expected by chance from the two coders' marginal label frequencies across the source character positions:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \quad p_e = \sum_{c \in \mathcal{C}} p_A(c) p_B(c). \quad (4)$$

Here,  $p_o$  is the fraction of source character positions with identical labels, and  $p_A(c)$  and  $p_B(c)$  are the two coders' label proportions. The label set  $\mathcal{C}$  comprises L/K/S/T and O. For the four-label scheme plus O, the available exports yield  $p_o = 0.771$  and  $p_e = 0.471$ , giving  $\kappa = 0.567$ . These character-level quantities are distinct from the span counts in Eq. (2); the 2,571 positions are nested within 50 sentences, not 2,571 independent samples.

Empty–empty sentences contribute no span true positives. Span matching and character-level agreement use different units; reproduction materials document optional category projections. The post-generation review sample instead compares a machine-prefilled layer with human review: 140 exact typed matches among 144 machine and 148 reviewed spans give P/R/F1 of 0.972/0.946/0.959. Ninety-five sentences have identical complete span sets. The sample was stratified by historical source, empty/nonempty status, and text length from 2,179 eligible records. The frame excludes 100 initially reviewed records, 100 records reserved for an earlier pilot, and 72 pending records from the 2,451-record original Silver pool; two small strata totalling 11 records were unsampled. The estimate describes this reviewed sample; it does not provide an unbiased estimate of accuracy across the corpus.

The 15-sentence assisted-review subset is nested within the post-generation Silver review cohort. Both annotators had access to machine coding suggestions. Their human layers contain the same 27 spans, yielding exact span F1 and character kappa of 1.000. These values describe agreement under machine-assisted review, where shared suggestions can influence both annotators; they are not evidence of independent blind coding or corpus-wide reliability under the final handbook.

**Historical annotation profile.** Median input lengths are 33 Unicode code points in each B2 supervision split, 41.5 in the reference, and 33.5 in the additional review sample. Median span length is six code points in supervision and both review layers, compared with four in the reference. Counts include spaces and punctuation; Figure 3 shows these historical distributions.

### Dual-Reviewer QA on the Expanded Silver Sample

The completed export contains both human layers for the 150 preselected sentences, with all confirmations recorded separately for each reviewer. Identifiers exactly match the stratified draw, and normalized sentence texts have no overlap with the original 350. The sample includes 60 cloud, 45 public-sector, and 45 listed-company sentences. The recorded Astra selection in Codex and Grok in Cursor supplied visible suggestions (Table 13); scores reflect assisted reviewer–reviewer agreement, not blind coding.

The 49 sentences with differing human span sets remain part of the raw agreement calculation. The exported annotations are not replaced by their intersection or a consensus layer. Character-level calculations assign O/L/K/S/T to original Unicode-code-point positions, including spaces



**Table 12.** Paired agreement in the additional QA150 sample. Exact typed span F1 uses original pre-adjudication span sets. Percentile 95% confidence intervals use 10,000 sentence resamples within source strata (seed 20260917), conditional on the source quotas.

| Comparison             | Exact F1 | 95% interval   | Identical sentences |
|------------------------|----------|----------------|---------------------|
| Reviewer A–Reviewer B  | 0.816    | [0.769, 0.860] | 101/150             |
| Astra–Grok suggestions | 0.962    | [0.937, 0.983] | 138/150             |
| Reviewer A–Astra       | 0.823    | [0.776, 0.866] | 100/150             |
| Reviewer B–Astra       | 0.850    | [0.803, 0.892] | 108/150             |

and punctuation. Two positions have conflicting types within one reviewer layer and are excluded from both character sequences, leaving  $N = 5,990$  positions. All original spans remain in span-level F1. These raw review layers are not substituted for Gold150; the affected record is identified in the accompanying agreement data. Final adjudication and incorporation into later training targets have not been verified. The reported coefficients therefore describe the preserved raw review layers. The resulting observed agreement over retained character positions is  $p_o = 0.934$ , with  $p_e = 0.439$  and  $\kappa = 0.883$ .

For nominal Krippendorff's  $\alpha$ , let  $m_c$  count assignments to category  $c$  across both reviewers, with  $\sum_c m_c = 2N$ . The observed and expected disagreements are

$$\alpha = 1 - \frac{D_o}{D_e}, \quad D_o = 1 - p_o, \quad D_e = 1 - \frac{\sum_c m_c(m_c - 1)}{2N(2N - 1)}. \quad (5)$$

Here  $D_o = 0.066$  and  $D_e = 0.560$ , giving  $\alpha = 0.883$ . The character coefficients and span F1 describe different units of agreement (Artstein and Poesio, 2008); the many O positions contribute to the character calculation. Agreement between two machine suggestion layers likewise measures concordance, not accuracy against an independent human reference. Export hashes, calculation code, and detailed diagnostics accompany the reproduction record.

**Independent blinded coding by three annotators.** The formal sample contains eight listed-company, 16 public-institution, and 26 Tianchi sentences. The research team reports random selection, with one sentence per recorded advertisement identifier. Each annotator completed familiarization and training before independent coding. The archived exports establish sample membership and source counts, but the complete candidate frame and sampling execution log were not included in this release. The original draw therefore cannot be independently reproduced from the release; source-stratified bootstrap resampling quantifies agreement uncertainty conditional on this sample. Five- and 15-sentence training sets are excluded from formal agreement. The analysis uses the latest separate exports after individual self-review with access to the handbook. The final-handbook study does not relabel the historical model reference.

The three coders in Table 3 produced 104/123/113 spans; exact typed matches numbered 90/92/96 for A–B/A–C/B–C. Pairwise F1 ranges are 0.762–0.805 for K, 0.778–0.869 for S, and 0.865–1.000 for T. All coders identified the same two L spans in one sentence. O accounts for 63.2–64.1% of character positions. The linked reproduction materials provide the individual annotations, per-type counts, and bootstrap code; none of the 10,000 resamples produced an undefined reported coefficient.

## APPENDIX C. SUPPLEMENTARY EVALUATION

This appendix documents the shared evaluation protocol, uncertainty and error diagnostics, and exploratory expanded-pool results. Scores and differences are calculated before rounding and displayed to three decimals; scientific notation preserves the sign of near-zero interval endpoints. The supporting records provide per-run results, scorer details, and earlier protocol comparisons. Those histories use different output formats and demonstration settings and therefore do not support controlled estimates of the effects of changing prompts.

### Shared-Guideline Evaluation against the Human Reference

The shared protocol evaluates six configurations on all 150 reference sentences, using V4.2.12-based general instructions and five schematic examples.

**Configuration and output records.** Table 13 distinguishes the original annotation teacher, inference baselines, and expansion channels, with their recorded identifiers and access configurations.

**Table 13.** Recorded model identifiers, access routes, and roles in annotation and evaluation.

| Role                        | Recorded identifier                                                  | Access and configuration                                                                                                                         |
|-----------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Original Silver annotation  | <code>gpt-6-astra</code>                                             | OpenAI Codex interface; recorded model selection for adopted Silver labels.                                                                      |
| GPT (API)                   | Request: <code>gpt-5.4</code> ; returned: <code>gpt-5.6-terra</code> | Temperature 0; output cap 2,048.                                                                                                                 |
| Claude (API)                | <code>claude-sonnet-4-5-20250929</code>                              | Temperature 0; output cap 2,048.                                                                                                                 |
| DeepSeek (API)              | <code>deepseek-v4-pro</code>                                         | Official DeepSeek API; Non-thinking: temperature 0, cap 2,048. Thinking: cap 8,192; temperature not specified.                                   |
| Kimi (API)                  | <code>kimi-k2.6</code>                                               | Official Moonshot API; Non-thinking; temperature 0.6; cap 2,048.                                                                                 |
| Qwen baseline / LoRA        | <code>Qwen2.5-14B-Instruct</code>                                    | Local weights; deterministic decoding; output cap 2,048. Original Hub commit not recorded; archived file hashes identify the supplied artifacts. |
| JobBERT-zh                  | Released initialization and B2 continuation                          | Local encoder and inherited CRF; initialization-selection history incomplete.                                                                    |
| Expansion comparison        | Codex / API-returned <code>gpt-6-astra</code>                        | Alternative API batch: 2,500 candidates, 13 September 2026, ten records per request. Earlier components are retained in both final pools.        |
| Assisted-review suggestions | <code>gpt-6-astra</code> ; Grok                                      | Codex and the Grok selection in Cursor supplied suggestions for QA150. The Grok version and complete decoding settings were not retained.        |

Baseline calls were completed on 10–11 September 2026. Claude returned the dated identifier shown above to a request for `claude-sonnet-4-5`. Codex and Cursor identify the annotation interfaces. GPT and Claude were accessed through a third-party API gateway and are named as returned by that service; those identities were not independently verified against vendor APIs. Retained prompts and parameters accompany the configuration files.

Models return an ID and spans as literal text, zero-based occurrence, and L/K/S/T type. The

span parser resolves spans to Unicode offsets for exact and relaxed scoring with the released span scorer. All supplied input texts match the frozen reference.

Qwen uses deterministic decoding; Kimi uses non-thinking mode at temperature 0.6. DeepSeek output caps are 2,048 tokens without thinking and 8,192 with thinking. Access routes appear in Table 13; full configurations accompany the reproduction notes.

**Parser and matching conventions.** The parser resolves literal spans to source offsets, sorts candidates by start offset and original output order, and keeps the first compatible candidate. Later duplicates, same-boundary type conflicts, and overlaps are rejected. Partial responses retain accepted spans; fully rejected or malformed responses become empty predictions. Identifier coverage is checked across all supplied sentences before scoring.

Exact matching deduplicates identical span tuples within each sentence. Relaxed matching traverses reference spans in source order and selects the unused same-type prediction with highest IoU, taking the first prediction on a tie. The threshold is 0.5, and matching is greedy rather than globally optimized. Empty–empty records contribute no span counts. The scorer specification preserves schema validation, BIO edge cases, and executable checks. The sentence-level counts reproduce the published exact scores before resampling.

**Conditional resampling.** To allow dependence among sentences with the same source identifier, we grouped the 150 reference sentences by the prefix preceding –s, giving 61 groups. Each of 10,000 paired bootstrap draws samples 61 complete groups with replacement and includes all their sentences. The same draw is used for all compared models and each fixed training seed; micro-F1 is recomputed before seed means and paired differences are calculated. Table 6 reports percentile intervals. The prefixes agree with the stored source-document identifiers in the original corpus. Group resampling accounts for dependence among sentences from the same document; it does not change split membership. These secondary intervals condition on the selected reference and trained models, and are not adjusted for multiple comparisons. The reproduction package retains the earlier sentence-bootstrap results and the seed used for the group analysis. Training-seed SD measures a separate source of variation. As a sensitivity check, widening the bootstrap intervals for the six displayed comparisons using Bonferroni tail probabilities leaves all inclusion/exclusion-of-zero conclusions unchanged; detailed intervals and code are in the supplementary calculations.

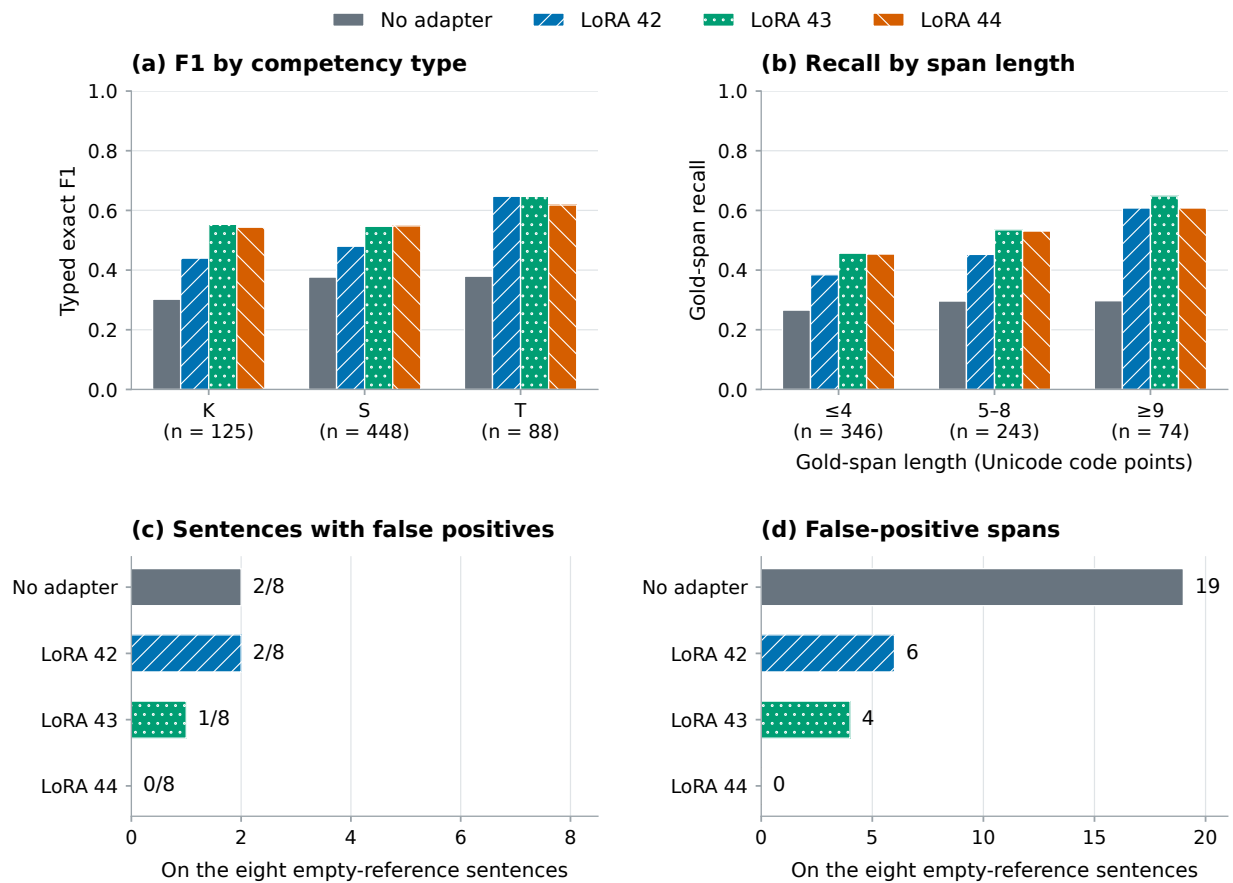
**Observed mismatch examples.** In the frozen no-adaptor Qwen output for 1801–s0004, the predicted S span [5,24) is 参与公司无人机飞控系统技术项目开发策划 (*participate in planning a technical development project for the company’s UAV flight-control system*), whereas an overlapping reference S span is [16,24), 技术项目开发策划 (*technical project development planning*). For 1802–s0005, both layers select 方法 (*method*) at [6,8), but the prediction is S and the reference is K. These frozen outputs illustrate boundary and type mismatches with the reference; they have not been separately adjudicated. After removing exact matches, the pairing procedure identifies 27 same-boundary/different-type pairs, 89 same-type/overlapping-boundary pairs, and 65 other unmatched predictions, totalling 181 false positives. These counts classify span relationships rather than linguistic causes. The error-analysis data records the pairing order and parser errors; a full adjudicated error taxonomy remains unavailable.

The released predictions and scoring code reproduce the reported exact and relaxed scores.

Under their respective output budgets, DeepSeek in thinking mode has higher precision and lower recall than the recorded non-thinking configuration.

**Per-seed Qwen results.** The per-seed record retains precision, recall, exact/relaxed F1, and calibration/challenge scores underlying Tables 7 and 6. Boundary exact F1 in Table 7 was subsequently recomputed from the same frozen accepted spans, with types collapsed and duplicate coordinates removed within sentences. This adds a scoring view without new training or inference. The model-diagnostic records provide per-seed and per-type precision, recall, TP/FP/FN, and the full-precision summaries used in Table 8; Figure 5 retains individual-seed Qwen results.

## Matched Qwen Frozen-Prediction Diagnostics



**Figure 5.** Qwen diagnostics on the same 150 reference sentences. (a) Typed exact F1 by competency type. (b) Recall by gold-span length;  $n$  counts gold spans. (c–d) False positives on eight empty-reference sentences, counted by sentence and span. Colour and pattern identify runs.

Diagnostics reuse the same 150 reference sentences and each run's frozen accepted spans. Typed exact matching is completed on each full sentence before aggregation by gold-span length. Bins ( $\leq 4$ ,  $5-8$ ,  $\geq 9$  Unicode code points) are fixed in advance; they contain 346/243/74 gold spans in 102/110/56 sentences, respectively. A sentence may contribute gold spans to multiple bins. Type-specific unmatched predictions count as FP even in empty-reference sentences. Results for each training seed are evaluated separately on the same reference sentences.

On the eight empty-reference sentences, accepted false-positive spans decrease from 19 without an adapter to 6, 4, and 0 for seeds 42, 43, and 44; affected sentences number 2, 2, 1, and 0. L has two gold spans: the no-adapter run recovers 0/2 and each LoRA run 2/2, with one extra L prediction in every run. Complete TP/FP/FN, legal-empty-output counts, and parser-rejection diagnostics are retained in the diagnostic records.

### Exploratory Expanded-Silver Experiments

**Expanded-pool adaptation.** The expanded Qwen runs use seed 42 and the shared prompt, occurrence output, parser, and Qwen training settings in Table 4. Loss applies only to assistant output. Both pooled checkpoints are selected at epoch one using Silver-validation exact F1; decoding is deterministic with a 2,048-token output cap. The complete configurations retain the parameter records. The expanded JobBERT runs retain the domain-adapted encoder and inherited CRF. Its latest run uses 8,842 training and 698 development records, so the separate 350-sentence Silver-test interpretation applies to Qwen, not to that JobBERT run.

Table 14 reports the two pooled Qwen runs against their held-out Silver targets and the shared human reference; the main results discuss the distinction between these evaluation targets.

**Table 14.** Expanded-pool Qwen results (seed 42). Silver columns measure agreement with generated targets; Gold150 columns use human-finalized labels. Scores are typed exact micro-F1 except the final relaxed-F1 column.

| Pool variant             | Silver validation agreement | Silver test agreement | Gold150 exact | Gold150 relaxed |
|--------------------------|-----------------------------|-----------------------|---------------|-----------------|
| Alt.: 8,943 / 351 / 352  | 0.783                       | 0.759                 | 0.597         | 0.713           |
| Codex: 8,842 / 348 / 350 | 0.833                       | 0.809                 | 0.585         | 0.703           |

Alt. is the supplementary alternative pool; Codex is adopted. Counts give train / validation / test sentences. Both retain earlier B2 and public-prompt annotations; names identify the channel for the replaced 2,500-sentence subset. Retained totals are 9,646 (Alt.) and 9,540 (Codex). Checkpoints were selected at epoch one with patience two. Differences use unrounded scores; rounding applies only to displayed values.

The Codex variant has higher agreement with its own Silver test labels but an approximately 0.011 lower exact F1 on Gold150 than the alternative variant. The channel replacement also changes record inclusion: 2,353 of the 2,500 Codex annotations are retained, compared with 2,459 alternative annotations. The contrast describes the resulting pools rather than an isolated annotation-channel effect.

The expanded JobBERT runs obtain Gold150 exact F1 of 0.242 for the alternative pool and 0.232 for the Codex pool (seed 42); the latter has relaxed F1 of 0.485. A preceding 7,117-record configuration gives  $0.237 \pm 0.002$  across three seeds. These scores are lower than the historical B2 continuation result, but the expanded runs change the annotation protocol, source composition, and development set alongside training size. The latest JobBERT run selects on 698 development records, whereas Qwen uses 348 validation records and retains 350 for Silver testing. Thus these results do not identify an effect of increasing training size alone. They show that the larger supervision pool does not deliver uniform improvements across the recorded baselines; the matched Qwen comparison remains the direct evidence for adaptation under a fixed inference protocol. For comparison, the B2 Qwen adapter at seed 42 scores 0.500 on Gold150; its three-seed mean of 0.540 is a different summary. Neither is a matched estimate of an expansion effect because

the expanded runs change the supervision pool and selection targets. The lower expanded Job-BERT scores remain unexplained because the available records do not isolate label-version, input-windowing, or decoding effects. These runs remain exploratory. Exploratory supervision-subset runs and their selection arrangements are preserved in the supporting experiment records. These single-seed runs are not controlled estimates of training-size or annotation-channel effects.

## APPENDIX D. DATA LINEAGE, ACCESS, AND REPRODUCTION

Table 17 identifies the released data, models, and scoring materials. The reproduction guide links their instructions. The supplementary audit package supplies the calculations below, input hashes, and executable checks. Earlier experiment outputs and prompts remain available in the repository; moving the external-task experiments out of this paper does not remove their records.

### Source Categories and Data Preparation

The four original collection labels describe acquisition batches; the three acquisition routes identify MacroData, public-institution collection, and Tianchi. They are different classifications. The original training partition contains 10,312 graduate-recruitment and 7,148 AI sentences; development contains 2,143 AI sentences; test contains 1,423 AI, 473 Tianchi/cloud, and 1,341 public-institution sentences. The later expansion uses Tianchi/cloud, public-institution, and listed-company source groups. The blinded agreement sample's 26/16/8 counts refer to these later source groups and cannot be read as proportions of the four original collections.

MacroData's operator is Chongqing Maherui Information Technology Co., Ltd. Public-institution records retain source names, titles, URLs, listed publication dates, and geographic fields where available. Publication dates are not collection dates; a complete crawl-window and source-exclusion history has not been recovered. The authors confirm academic-research use permission. Repository notices separately identify the rights in author-created annotations and the restrictions on third-party text.

B2's 2,319 training/development records contain 2,079 distinct NFC-normalized complete texts. Removing whitespace and trimming edge punctuation gives 2,075 preparation keys; eleven fail the length or punctuation filter, leaving 2,064. Thus the difference of 255 is not entirely duplicated text. These counts describe preparation, not a new deduplicated training experiment.

The preparatory target combined those 2,064 valid keys with 7,936 new candidates. Final assembly used different admission rules and retained 2,077 earlier B2 sentences. The new candidates were processed as batches of 3,620, 1,816, and 2,500. Table 15 reconciles the final totals; exclusions of 213 and 113 in the first two listed batches, offset by 13 additional earlier records, explain the previously obscured difference of 313. The supplementary alternative differs in the retained channel-comparison batch. Counts do not establish a semantic explanation for every excluded annotation.

New-candidate selection used requirement-related keywords, fixed hash ranking, and recorded source quotas. HTML/whitespace cleaning, sentence splitting, and length filters preceded annotation. Exclusion checks used the stored sentence strings and available pretraining dump. The 3.2-million-line dump's content hash was not retained, so the historical pretraining exposure cannot be fully reconstructed. The original frozen labels remain tied to their generation-time handbook; one later rule-compatible example cannot certify an entire historical pool.

**Table 15.** Final expanded-pool composition. Counts are retained sentences, separately reconstructed from the frozen split files.

| Component                | Candidates     | Codex retained | Alternative retained |
|--------------------------|----------------|----------------|----------------------|
| Earlier B2 component     | Stage-specific | 2,077          | 2,077                |
| First two new batches    | 3,620          | 3,407          | 3,407                |
| Third new batch          | 1,816          | 1,703          | 1,703                |
| Channel-comparison batch | 2,500          | 2,353          | 2,459                |
| Final pool               | —              | 9,540          | 9,646                |

## Split Membership and Document Overlap

Original records are grouped by their stored source-document identifier; new expansion records use source-file and source-row identifiers. Every inspected record maps within its source namespace. No crosswalk between the old and new namespaces is available, so the observed overlaps are not an exhaustive cross-source duplicate analysis.

**Table 16.** Shared source-document groups in the later experimental splits. Counts refer to different sentences within shared groups, not identical sentences across splits.

| Comparison                              | Shared groups | Left sentences | Right sentences |
|-----------------------------------------|---------------|----------------|-----------------|
| B2 train / development                  | 1             | 4              | 27              |
| B2 train / Gold150                      | 59            | 531            | 143             |
| B2 development / Gold150                | 0             | 0              | 0               |
| Expanded Codex train / validation       | 185           | 1,060          | 208             |
| Expanded Codex train / test             | 188           | 1,076          | 210             |
| Expanded alternative train / validation | 186           | 1,069          | 210             |
| Expanded alternative train / test       | 189           | 1,079          | 211             |

No exact or NFC-normalized sentence is shared between the later B2 training/development splits or between training and Gold150. NFKC normalization with whitespace removal and case folding finds two B2 training/development matches and one training/validation match in each expanded variant. No normalized training/test sentence matches were found in the expanded variants. These sentence checks do not remove the shared source-document groups in Table 16; all original predictions and scores are retained.

The original 22,840-sentence split already enforced source-document separation, as documented by the earlier split audit and reproduced from the frozen corpus. The later B2 manifest explicitly specifies sentence-level isolation, and its training, development, and Gold150 file hashes match the evaluated Qwen package. The additional group analysis concerns those later assembled splits; it does not invalidate the original split or the successful reference-sentence exclusion checks. B2 training contains 1,910 distinct complete texts among 2,150 records. Repeated records remain weighted in training. No new deduplicated training run is implied by this audit, and shared source documents alone do not quantify any effect on model scores.

**Agreement-sample membership.** No agreement sentence matches B2 training/development (2,150/169) or Gold150 by ID or normalized complete text. Each expanded variant contains 46 in training, two in validation, and one in its Silver test; one is absent. Human coding remains

blinded, but adjudication cannot remove this exposure from already trained models. The membership audit records the frozen-input hashes and counts.

## Resource Access and Privacy Screening

**Table 17.** Component access and the reproduction level supported by the identified releases. Local possession is distinguished from public access.

| Component                     | Access route / available objects                                                                      | Scope and limitation                                                                                                                                                                       |
|-------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Core v0.1.3                   | Dataset archive: corpus, Gold150 (150), B2 (2,150/169), texts and labels                              | Download the named version; inspect labels and use its splits. It does not contain the later expanded pools.                                                                               |
| Qwen reproduction archive     | Model archive: three B2 adapters, selected expanded adapters, frozen predictions, prompts and scorer  | Follow its model index for inference or rescoring; obtain base weights separately. Full training inputs are not bundled.                                                                   |
| Expanded pools                | 9,540 adopted / 9,646 supplementary records; complete files verified locally against split manifests  | IDs and selected outputs are archived; public complete texts and labels, or an executable controlled-access arrangement, remain unavailable. Not publicly retrainable as a complete study. |
| QA150 and historical coding   | Pseudonymized texts and human layers linked in Data Availability; source hashes and cohort membership | Recalculate agreement without account metadata. The release preserves design distinctions and does not establish final-rule blind reliability.                                             |
| Three-annotator blinded study | Agreement release: 50 sentences, three individual label layers, source IDs, scorer and intervals      | Recalculate independent agreement; five + 15 training sentences excluded. This is not a model test set.                                                                                    |
| Chinese encoder baselines     | Six public Hugging Face checkpoints; audit DOI 10.5281/zenodo.22937245                                | Frozen predictions, code and configurations support rescoring. Local rescoring verified all six runs; server checkpoint re-inference is separately recorded.                               |

Reuse terms vary by component. The authors' annotation permission excludes third-party text; separate model/base-model notices apply. Access on request and exact reconstruction from live recruitment pages are not guaranteed.

An automated screen examined text fields in 15 public corpus and annotation files, covering 28,635 distinct complete texts. Email, mobile-number, and landline patterns flagged 38, 74, and 50 texts, respectively; categories overlap. No such candidates were found in Gold150 or QA150, but candidates occur in B2 and in the separate agreement sample. These are pattern matches requiring human review, including possible organizational contact details or false positives. The screen does not certify anonymization, cover all names or personal information, or replace review of redistribution permissions. The public audit reports aggregate counts; the masked record-level review list is retained locally for resolution before a privacy-vetted release.

**Historical model implementation.** JobBERT-zh is the project's released Chinese recruitment-domain encoder, with an inherited CRF used in the reported continuation runs. The exact original tokenizer revision, some pretraining/selection settings, and truncation or windowing counts were



not recovered. Its 256-token limit is not a character limit; the reference retains all target spans. These gaps constrain reconstruction of historical training, although released predictions can be rescored.

**Manuscript preparation.** OpenAI Codex assisted with manuscript revision, LaTeX formatting, and code preparation and checking under author direction. Preparation tools were used in September–October 2026; their model versions were not retained. The current workflow diagrams are vector drawings with editable source files.

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## COMPETING INTERESTS

The authors declare that they have no competing interests.

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